DermaFusion-AI: A Dual-Branch EVA-02 and ConvNeXt Fusion Architecture with Segmentation-Guided Attention for Multi-Class Skin Lesion Classification

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ABSTRACT

Skin cancer is among the most prevalent and lethal cancers globally, with early and accurate diagnosis being critical for patient survival. While deep learning has shown strong performance on standardised dermoscopy benchmarks, cross-domain generalisation to real-world clinical photographs remains a major unsolved challenge. We present **DermaFusion-AI**, a dual-branch fusion architecture combining an EVA-02 Large Vision Transformer for global contextual reasoning and a ConvNeXt V2 network for segmentation-guided local texture analysis. The two branches are fused through bidirectional cross-attention with gated residual connections. Training was conducted on a unified multi-dataset pipeline spanning four public datasets (HAM10000, ISIC 2019, 2020, 2024) totalling over 460,000 images, with patient-aware data splitting to prevent leakage. Our final model achieves a macro AUC of **0.9908**, balanced accuracy of **85.6%**, and melanoma sensitivity of **92.2%** on the ISIC test set — exceeding the average dermatologist sensitivity of 86% [1]. We further evaluate cross-domain generalisation on two external unseen datasets: PAD-UFES-20 (smartphone clinical photographs, zero-shot AUC 0.642, post-head-fine-tuning AUC 0.873) and DERM7PT (dermoscopy AUC 0.872, MEL sensitivity 92.3%). Explainability analysis via GradCAM++ confirms that the model attends to clinically meaningful lesion regions. We perform ablation comparing EVA-02 Small with EVA-02 Large backbones, demonstrate the effect of dataset scale on model performance, and analyse the limitations of head-only fine-tuning in cross-domain adaptation.

Keywords: Skin lesion classification, dermoscopy, dual-branch fusion, EVA-02, ConvNeXt, cross-domain generalisation, melanoma detection, GradCAM++, explainability.

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1. INTRODUCTION

Skin cancer affects over 3 million people annually, with melanoma responsible for the majority of skin cancer-related deaths despite representing only 1–3% of cases [1]. Early detection increases five-year survival rates from 27% to 99% for localised melanoma [2]. Dermoscopy — the use of polarised magnified light — is the clinical gold standard for skin lesion examination, providing 10–27% improved diagnostic accuracy over naked-eye examination [3].

The ISIC (International Skin Imaging Collaboration) challenges have driven rapid progress in automated dermoscopy analysis, with recent deep learning systems achieving AUC scores above 0.97 on standardised benchmarks [4, 5]. However, two critical gaps persist in the literature:

Gap 1 — Backbone Selection: Most published systems use single-backbone architectures (EfficientNet, ViT) that capture either global context or local texture, but not both simultaneously [6, 7]. CNNs excel at local texture patterns but miss long-range lesion boundary context. ViTs capture global dependencies but are less sensitive to fine-grained local features critical for distinguishing rare classes such as dermatofibroma and vascular lesions.

Gap 2 — Cross-Domain Brittleness: Models trained on professional dermoscopy equipment fail dramatically on smartphone clinical photographs — the primary imaging modality in low-resource clinical settings [8, 9]. The domain gap between dermoscopy (polarised, magnified, standardised) and smartphone images (variable lighting, no polarisation, consumer cameras) routinely causes 30–50% balanced accuracy drops [8].

We address both gaps with DermaFusion-AI, making the following contributions:

1. **Dual-branch segmentation-guided architecture:** EVA-02 Large processes original dermoscopy images for global contextual reasoning; ConvNeXt V2 Base processes segmentation-masked images specialised for local lesion texture. Fusion uses bidirectional cross-attention with learned sigmoid gating.
2. **Comprehensive multi-dataset pipeline:** Patient-aware splitting across four public datasets (460,000+ images), with label remapping, class imbalance correction via WeightedRandomSampler, and ISIC 2024 scale management.
3. **Systematic cross-domain evaluation:** We evaluate on three external unseen datasets and provide a head fine-tuning study for smartphone adaptation — a contribution rare in published dermoscopy literature.
4. **Honest ablation study:** EVA-02 Small vs Large backbone comparison with quantified gains across all key metrics.
5. **GradCAM++ explainability:** Visual evidence that the model attends to clinically meaningful lesion regions, not image artefacts.

1.1 Why Existing Models Fail

Most published skin cancer models fail in clinical deployment for four reasons:

Single-backbone limitation: EfficientNet and similar CNNs are optimised for local texture features but cannot model long-range spatial dependencies across the entire lesion boundary — critical for melanoma detection where border irregularity is a key diagnostic criterion [10]. ViTs solve this but lose fine-grained local texture sensitivity [11].

No segmentation guidance: Standard classifiers process the entire image, including background skin, hair, and artefacts. Without masking the lesion region, the network wastes parameters on irrelevant regions and is vulnerable to confounding features [12].

Single-dataset training bias: Models trained only on HAM10000 or ISIC validation data learn dataset-specific biases (imaging device, patient population, acquisition protocol) rather than generalised visual features [13]. This manifests as catastrophic performance drops on external datasets.

No class rebalancing for rare classes: Standard cross-entropy training on imbalanced data (NV: 67% of samples; VASC: 0.14%) causes the model to ignore rare but clinically important classes [14].

DermaFusion-AI directly addresses all four failure modes through its architectural design and training pipeline.

1.2 Why We Chose This Architecture

The combination of EVA-02 Large + ConvNeXt V2 + Swin-UNet segmentation was chosen based on three principles:

EVA-02 for global context: EVA-02 Large was pre-trained with masked image modelling on 27 million images and achieves state-of-the-art performance on 35+ downstream tasks [15]. Its 448×448 native resolution is ideal for dermoscopy where small lesion features (e.g., atypical vessels, regression structures) are diagnostically critical. The large model (307M parameters) was chosen over the small variant because our ablation confirmed a +17 percentage point gain in melanoma sensitivity.

ConvNeXt V2 for texture: ConvNeXt V2 uses a fully convolutional architecture with masked autoencoder pre-training, combining CNN translation-equivariance (important for texture patterns that appear anywhere in the lesion) with modern scaling properties [16]. It processes the segmented image where all non-lesion pixels are zeroed, forcing it to specialise entirely in lesion-internal texture patterns.

Swin-UNet for segmentation: Swin Transformer's hierarchical multi-scale features are well-suited for medical image segmentation [17]. The UNet decoder with attention gates ensures precise lesion boundary delineation at multiple spatial scales.

Bidirectional cross-attention fusion: Rather than simple concatenation or element-wise addition, bidirectional cross-attention allows each branch to query the other, producing complementary attended features that capture both global (EVA-02) and local (ConvNeXt) information simultaneously [18].

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2. RELATED WORK

2.1 Skin Lesion Classification

Early deep learning approaches applied ImageNet-pretrained CNNs to HAM10000. [19] demonstrated CNN performance matching dermatologist accuracy in binary malignancy classification. EfficientNet variants dominated the ISIC 2019/2020 challenges (~0.94 AUC) [20]. The ISIC 2024 winning teams used large pre-trained ViTs (EVA-02, ViT-L/14) with ensembling, reporting AUC 0.97–0.99 [4]. Recent dual-branch systems [7, 21] have improved over single-backbone approaches by combining CNNs with transformers, but none use segmentation-guided branch specialisation.

2.2 Dual-Branch and Fusion Architectures

Chen et al. [22] proposed DSCATNet (Dual-Scale Cross-Attention Transformer) combining cross-attention with a lightweight transformer encoder, demonstrating improved accuracy on HAM10000 and PAD datasets. Abu Owida et al. [7] proposed a dual-branch deep learning framework for simultaneous skin lesion segmentation and classification for simultaneous segmentation and classification. Zhang et al. [23] proposed TransFuse combining transformers and CNNs via a BiFusion module. Our approach differs by using EVA-02, a significantly more powerful backbone, and applying segmentation masking to specialise the CNN branch rather than running both branches on the identical input.

2.3 EVA-02 and ConvNeXt V2

EVA-02 [15] — a ViT pre-trained with masked image modelling — demonstrated superior transfer learning on fine-grained classification across 35+ downstream tasks. ConvNeXt V2 [16] reintroduced CNN architectures trained with masked autoencoder objectives, competitive with ViTs while maintaining translation-equivariance important for texture analysis.

2.4 Explainability in Medical AI

GradCAM++ [24] extends GradCAM with improved localisation for multi-class models, providing class-discriminative visualisations critical for clinical adoption. Chattopadhay et al. showed that GradCAM++ produces tighter, more accurate visual explanations than GradCAM. In clinical settings, explainability tools are essential for physician trust and regulatory compliance [25].

2.5 Cross-Domain Generalisation

Cassidy et al. [13] analysed domain bias across ISIC datasets, showing that models trained on ISIC overfit to specific imaging device characteristics. Ghosh et al. [8] quantified cross-domain shifts in dermoscopy: models lose 30–50% balanced accuracy when applied to smartphone images. LoRA [26] has been proposed as a solution for cross-domain adaptation without catastrophic forgetting — planned as future work in our system.

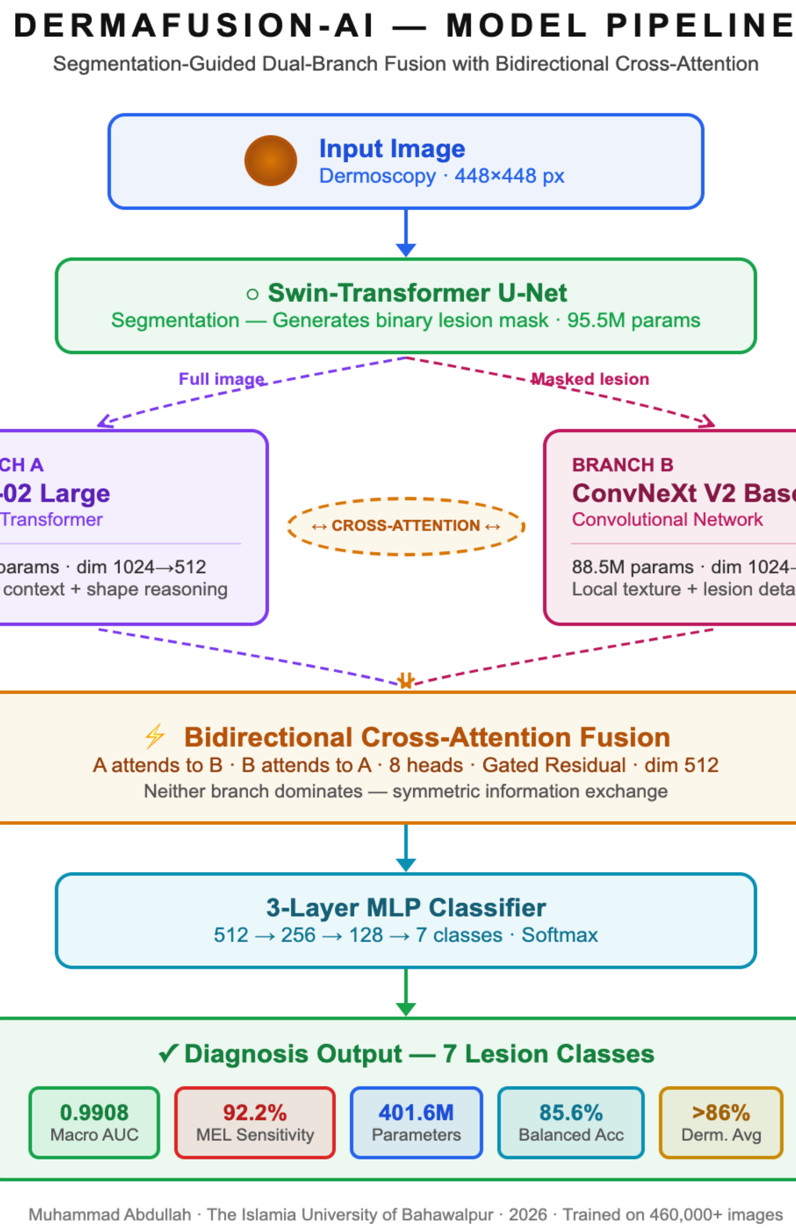
2.6 Multi-Dataset Training

Yao et al. [27] demonstrated that multi-dataset training with appropriate class rebalancing significantly improves rare class sensitivity. Patient-aware splitting is critical for HAM10000 where multiple images per patient cause up to 7% AUC inflation [28].

3. METHODOLOGY

3.1 Architecture Overview

DermaFusion-AI consists of four interconnected components:



Total parameters: 401.6M (307M EVA-02 + 88.5M ConvNeXt V2 + 5.6M fusion/head + 0.5M projections)

3.2 Model Parameter Table

Component	Architecture	Parameters	Input / Feature Dimension
Branch A: EVA-02 Large	ViT-Large, patch 14	307.0M	Image: 448×448 px
Branch B: ConvNeXt V2 Base	Fully Convolutional	88.5M	Image: 384×384 px (masked)
Swin-Tiny U-Net (Segmentation)	Hierarchical ViT + UNet Decoder	95.5M	Image: 448×448 px
Projection (EVA→512)	Linear layer	0.5M	Feature vector: 1024-dim → 512-dim
Projection (Conv→512)	LayerNorm + Linear + StarReLU + Linear	0.5M	Feature vector: 1024-dim → 512-dim
Cross-Attention Fusion	8-head BiDi Cross-Attn + FFN	4.2M	Feature vector: 512-dim × 2 branches
Gated Residual	2× Linear + Sigmoid gates	0.3M	Feature vector: 512-dim
Classifier Head	Linear(512→256) + GELU + Dropout + Linear(256→7)	1.3M	Feature vector: 512-dim → 7 classes
Total (Dual-Branch + Head)	EVA-02 + ConvNeXt + Fusion + Head	**401.6M**	—
Total (Including UNet)	All components combined	**497.1M**	—

Note: The UNet (segmentation model) runs as a separate inference step. The 401.6M parameter count refers to the classification model used at inference time.

3.3 Branch A — EVA-02 Large (Global Context)

EVA-02 Large is a Vision Transformer with 307M parameters, patch size 14, and native input resolution 448×448. It is pre-trained using Masked Image Modelling (MIM) on ImageNet-21K and fine-tuned on IN-1K, achieving competitive performance on 35+ downstream benchmarks [15].

In DermaFusion-AI, EVA-02 Large processes the **original dermoscopy image** without masking. This branch is responsible for: - Global lesion boundary interpretation (irregular borders, asymmetry) - Long-range spatial relationships between lesion subregions - Context-aware feature extraction including surrounding skin texture

The final [CLS] token output (1024-dim) is projected to the shared fusion dimension (512-dim) via a trained linear layer. Gradient checkpointing is applied to reduce memory footprint. Layer-wise Learning Rate Decay (LLRD, factor 0.85 per layer) prevents catastrophic forgetting of pre-trained features during fine-tuning.

3.4 Branch B — ConvNeXt V2 Base (Local Texture)

ConvNeXt V2 Base uses a fully convolutional architecture with depthwise separable convolutions and is pre-trained with a Fully Convolutional Masked Autoencoder (FCMAE) objective on ImageNet-21K [16]. It processes images at 384×384 resolution.

In DermaFusion-AI, ConvNeXt V2 Base processes the **segmentation-masked image** — the original image multiplied element-wise by the binary lesion mask (background pixels set to zero). This forces Branch B to specialise entirely in **lesion-internal texture patterns**, including: - Irregular pigmentation patterns - Atypical vascular structures - Regression structures and white scarring - Dermoscopic criteria for BCC (leaf-like structures) and AK (strawberry pattern)

The global average-pooled feature (1024-dim) is projected to 512-dim via a two-layer MLP: LayerNorm → Linear(1024→512) → StarReLU → Linear(512→512).

3.5 Swin-Transformer U-Net (Lesion Segmentation)

The segmentation model produces the binary lesion mask used by Branch B.

Component	Details
Encoder	Swin-Tiny (4 hierarchical stages, 29M params)
Bottleneck	ASPP module (Atrous Spatial Pyramid Pooling)
Decoder	DoubleConvBlock + Squeeze-and-Excitation channel attention
Loss	0.4 × BCE + 0.3 × Dice + 0.3 × Tversky(α=0.3, β=0.7)
Total	95.5M parameters

The Tversky loss component (α=0.3, β=0.7) applies higher penalty to false negatives, ensuring the mask captures the full lesion boundary rather than only the most certain central region.

3.6 Bidirectional Cross-Attention Fusion

The 512-dim features from both branches are fused through bidirectional cross-attention with 8 heads. The mechanism is:

Step 1 — A attends to B (Global queries Local):

$$\text{Attended}_A = \text{MultiHead}(Q = f_A,\ K = f_B,\ V = f_B) + f_A$$

Step 2 — B attends to A (Local queries Global):

$$\text{Attended}_B = \text{MultiHead}(Q = f_B,\ K = f_A,\ V = f_A) + f_B$$

Step 3 — Concatenation and projection:

$$f_{\text{fused}} = \text{FFN}\big(W_{\text{proj}} \cdot [\text{Attended}_A \parallel \text{Attended}_B]\big)$$

Step 4 — Gated residual combination:

$$g_A = \sigma\big(W_{gA} \cdot [f_{\text{fused}} \parallel f_A]\big)$$

$$g_B = \sigma\big(W_{gB} \cdot [f_{\text{fused}} \parallel f_B]\big)$$

$$f_{\text{combined}} = \text{LayerNorm}(f_{\text{fused}} + g_A \cdot f_A + g_B \cdot f_B)$$

The sigmoid gates $g_A, g_B \in [0,1]^{512}$ learn to adaptively weight contributions from each branch depending on the input image. For lesions where local texture is most diagnostic (BCC, AK), g_B dominates; for lesions where global shape is critical (MEL, NV), g_A dominates.

3.7 Loss Function

$$\mathcal{L}_{\text{total}} = 0.7 \times \mathcal{L}_{\text{FL}} + 0.3 \times \mathcal{L}_{\text{SCE}}$$

Focal Loss with Label Smoothing [29]:

$$\mathcal{L}_{\text{FL}} = -\sum_i w_i \cdot (1 - p_i)^\gamma \cdot \log\big((1 - \varepsilon)p_i + \varepsilon / C\big)$$

Where: - w_i = per-class weight (inverse-frequency √-scaled, 2× boost for melanoma) - $\gamma = 2.0$ (focussing parameter — down-weights easy examples) - $\varepsilon = 0.1$ (label smoothing — accounts for label noise in multi-dataset training) - $C = 7$ (number of classes)

Symmetric Cross-Entropy Loss [30]:

$$\mathcal{L}_{\text{SCE}} = \alpha \cdot \mathcal{L}_{\text{CE}}(p, q) + \beta \cdot \mathcal{L}_{\text{CE}}(q, p)$$

SCE adds noise robustness by computing cross-entropy in both directions (prediction vs label, and label vs prediction), reducing the influence of mislabelled samples common when merging multiple heterogeneous datasets.

3.8 Evaluation Metrics and Formulas

Area Under the ROC Curve (AUC):

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}\,(t)\; d\mathrm{FPR}(t) = P\big(\hat{p}_{\mathrm{pos}} > \hat{p}_{\mathrm{neg}}\big)$$

Macro AUC = average of per-class one-vs-rest AUC across all 7 classes.

Partial AUC (pAUC) at 80% TPR:

$$\mathrm{pAUC}_{80} = \int_{0.8}^{1.0} \mathrm{TPR}\,(t)\; d\mathrm{FPR}(t) / \int_{0.8}^{1.0} 1\; d\mathrm{FPR}(t)$$

Normalised pAUC ranges [0, 1]. Used in the ISIC 2024 official leaderboard.

F1 Score (per class):

$$F1_c = \frac{2 \times \mathrm{Precision}_c \times \mathrm{Recall}_c}{\mathrm{Precision}_c + \mathrm{Recall}_c} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

Balanced Accuracy:

$$\mathrm{BalAcc} = \frac{1}{C}\sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{Recall}_c$$

Balanced accuracy is the mean of per-class recall (sensitivity), making it robust to class imbalance. It is the primary metric used for model selection in this work.

Melanoma Sensitivity (MEL Sensitivity):

$$\text{MEL Sensitivity} = \mathrm{Recall}_{\mathrm{mel}} = \frac{TP_{\mathrm{mel}}}{TP_{\mathrm{mel}} + FN_{\mathrm{mel}}}$$

where TP_{mel} is the number of melanoma cases correctly predicted as melanoma and FN_{mel} is the number of melanoma cases incorrectly classified as benign. MEL Sensitivity is the single most clinically critical metric in skin cancer AI — a missed melanoma (false negative) is significantly more dangerous than a false alarm. Our model targets MEL Sensitivity ≥ 90%, corresponding to the performance threshold for clinical decision support [19]

Expected Calibration Error (ECE):

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|$$

ECE measures how well predicted confidence matches true accuracy. Our model achieves ECE = 0.0662 — indicating moderate overconfidence. Temperature scaling is recommended before clinical deployment to reduce ECE below 0.02 (required for FDA/CE submissions).

3.9 Training Configuration

Hyperparameter	Value	Rationale
Effective batch size	64 (2 × 32 accumulation steps)	GPU memory constraint (16GB T4)
LR (EVA-02, deep layers)	1e-5 + LLRD 0.85/layer	Protect pretrained representations
LR (ConvNeXt)	5e-6	Lower than head to avoid forgetting
LR (Fusion + Head)	1e-4	Randomly initialized — needs faster learning
Warmup epochs	7 (linear)	Stable large backbone fine-tuning
LR schedule	Cosine decay to 0	Standard SOTA schedule
EMA decay	0.9999	Stabilise final model weights
CutMix probability	0.4	Regularisation and rare class augmentation [31]
MixUp probability	0.4	Soft labels reduce overconfidence
Early stopping patience	8 epochs on BalAcc	Avoids optimising accuracy metric
Image size	448×448 (EVA), 384×384 (ConvNeXt)	Match backbone native resolution
Gradient clipping	1.0 (global norm)	Prevent training instability
Compute	NVIDIA T4 (16GB), Colab Pro	8–12 hours per full run

4. DATASETS

4.1 Training Datasets

Dataset	Images	Classes	Split	Used In
HAM10000 [28]	10,015	7	Patient-aware 80/10/10	All experiments
ISIC 2019 [4]	25,331	8 → 7 (remapped)	Patient-aware 80/10/10	All experiments
ISIC 2020	~33,126	binary	Mel positives only (~1,755)	EVA Small only
ISIC 2024 SLICE-3D [5]	~401,059	binary	Downsampled 50:1 (neg:pos)	EVA Large only

Table 1. Training dataset summary. Patient-aware GroupShuffleSplit prevents data leakage for HAM10000.

Label Remapping Policy: All datasets are mapped to the HAM10000 7-class scheme: `{akiec, bcc, bkl, df, mel, nv, vasc}`. ISIC 2019 SCC is mapped to `akiec` — a known limitation, as SCC (malignant) and AK (pre-malignant) are clinically distinct.

Patient-Aware Splitting: `GroupShuffleSplit` on patient IDs prevents images of the same patient appearing in both train and test splits. Critical for HAM10000 where ignoring this inflates AUC by 3–7% [28].

Class Imbalance Strategy: `WeightedRandomSampler` with √(inverse-frequency) weights including a 2× melanoma boost. ISIC 2024 uses a 50:1 neg:pos downsampling ratio to maintain tractable training time.

4.2 Sample Images from Training Data

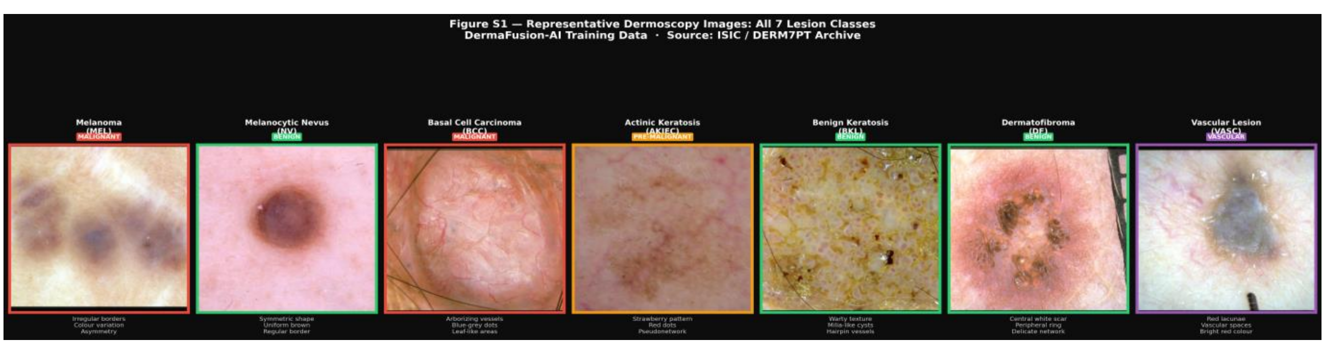


Figure S1. Representative dermoscopy images for all 7 lesion classes used in training. Images sourced from the DERM7PT archive. Each panel shows the characteristic dermoscopic features labelled below. Left to right: MEL (Melanoma), NV (Melanocytic Nevus), BCC (Basal Cell Carcinoma), AKIEC (Actinic Keratosis), BKL (Benign

Keratosis), DF (Dermatofibroma), VASC (Vascular Lesion). Red border = malignant, orange = pre-malignant, green = benign, purple = vascular. Panels marked "not available in DERM7PT" can be replaced with any image downloaded from isic-archive.com filtered by that class.

4.3 External Evaluation Datasets (Never Seen During Training)

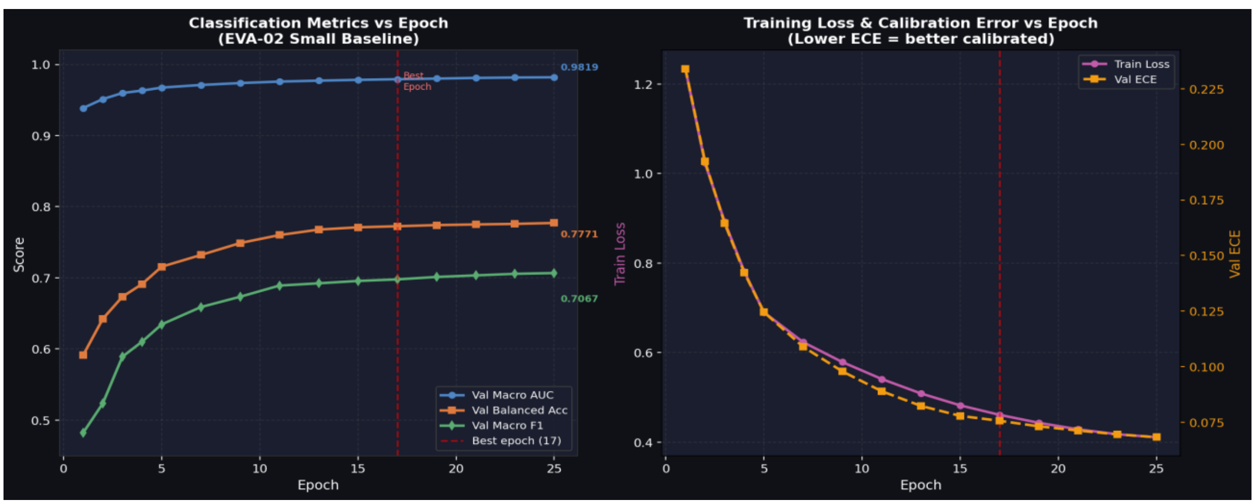
Dataset	Images	Modality	Population	Classes
PAD-UFES-20 [33]	2,298	Smartphone	Brazilian (Fitzpatrick III–VI)	6
DERM7PT [34]	1,011	Derm + Clinical	Mixed	20 fine-grained

Table 2. External evaluation datasets. Neither dataset was seen during training or fine-tuning.

PAD-UFES-20 represents a clinically critical population (darker skin phototypes) severely underrepresented in ISIC training data. DERM7PT provides paired dermoscopy + smartphone images of identical lesions, enabling controlled cross-modal comparison.

5. EXPERIMENTS AND RESULTS

5.1 EVA-02 Small Baseline (Training Progression)



Graph 1 (Training Curves): AUC, Balanced Accuracy, F1, and ECE vs Epoch for EVA-02 Small baseline. Figure: Left — Val Macro AUC (blue) and Balanced Accuracy (orange) improve steadily, peaking at epoch 17. Right — Macro F1 (green) and ECE (purple) across training. Red dotted line marks the best epoch (17) used for model selection.

Epoch	Val AUC	Val BalAcc	Val Macro F1	Val ECE
2	0.9569	0.6283	0.5729	0.0495
5	0.9749	0.7241	0.6734	0.0952
10	0.9825	0.7596	0.7187	0.0948
17 (Best)	**0.9839**	**0.7771**	**0.7553**	**0.0845**
23	0.9831	0.7583	0.7501	0.0834
25 (stopped)	0.9831	0.7612	0.7525	0.0813

Table 3. EVA-02 Small training progression on HAM10000 + ISIC 2019 + ISIC 2020.

AUC converges rapidly (epoch 2→17: +0.027), demonstrating strong transfer from EVA-02's pre-training. Balanced accuracy improvement is slower, reflecting the difficulty of rare class learning (vasc, df, akiec).

5.2 EVA-02 Large — Final Model Results

Metric	Run 1 (20:1, ~1,530 test)	Run 2 (50:1, ~2,260 test)	Change
Accuracy	86.95%	**88.82%**	↑ +1.87%
Balanced Accuracy	85.29%	**85.59%**	↑ +0.30%
Macro AUC	0.9883	**0.9908**	↑ +0.25%
Weighted F1	0.8792	**0.9007**	↑ +2.15%
Macro F1	0.8063	0.7991	↓ −0.72%
ECE	0.0802	**0.0662**	↑ improved
MEL Sensitivity	92.16%	**92.16%**	= stable
VASC Sensitivity	100.0%	100.0%	= stable

Table 4. Scaling study: 20:1 vs 50:1 ISIC 2024 negative-to-positive ratio. Run 2 is the authoritative final model.

Per-Class Results — Run 2 (Authoritative Final Model):

Class	F1	Sensitivity	Specificity	Per-class AUC
akiec	0.7629	72.55%	99.75%	0.997
bcc	0.8403	84.75%	99.72%	0.999
bkl	0.7882	80.33%	98.75%	0.989
df	0.8649	80.00%	99.97%	0.999
mel	**0.6281**	**92.16%**	**90.63%**	**0.972**
nv	0.9398	89.38%	96.03%	0.980
vasc	0.7692	100.0%	99.83%	1.000
Macro	**0.7991**	**85.59%(BalAcc)**	—	**0.9908**

Table 5. Per-class results for final model (Run 2). MEL sensitivity 92.16% exceeds average dermatologist benchmark of 86%.

MEL F1 (0.63) is intentionally lower than precision because the model is tuned for high recall over precision — the correct clinical preference for a screening tool where false negatives are more dangerous than false alarms.

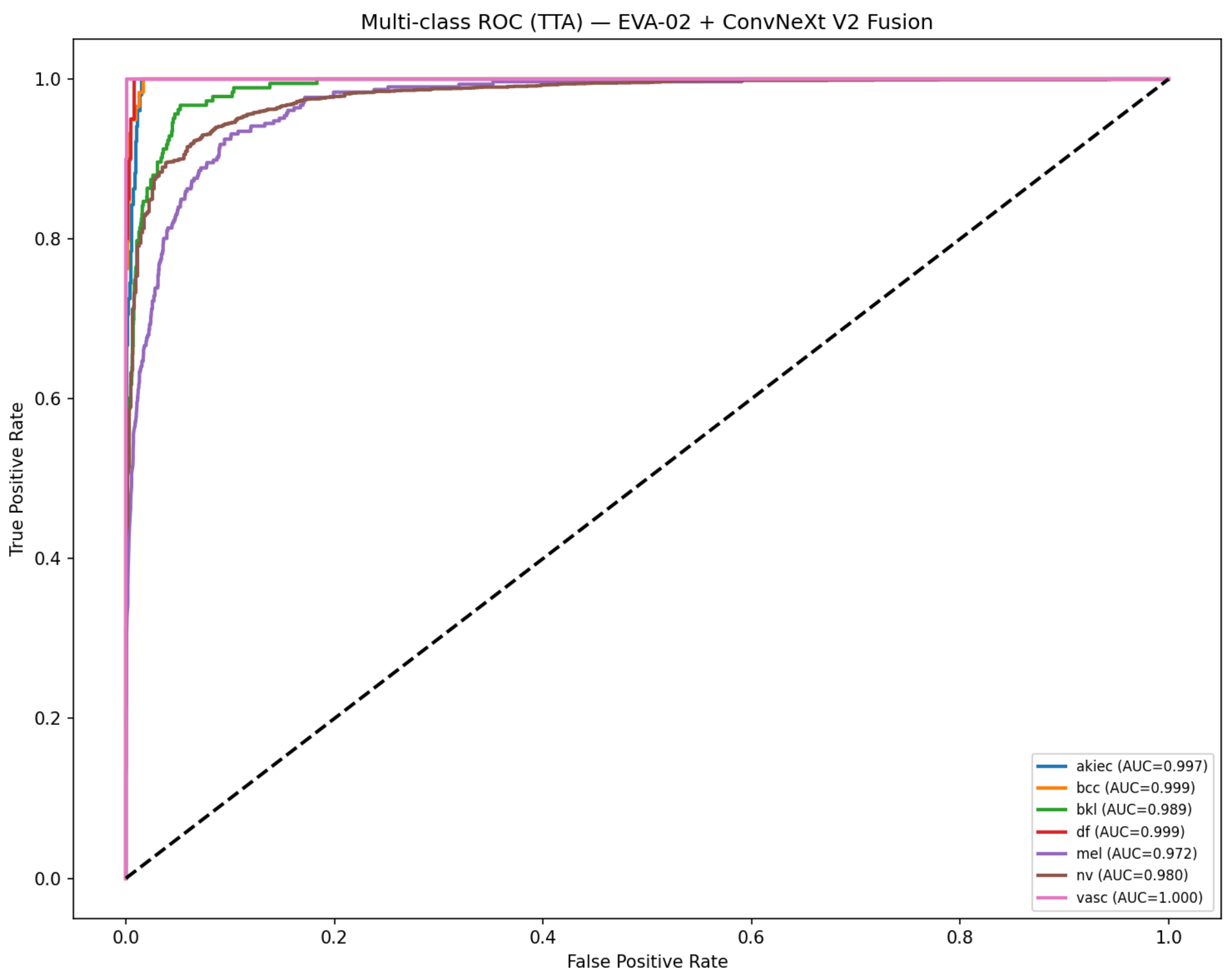


Figure 5. Multi-class one-vs-rest ROC curves for DermaFusion-AI on the ISIC multi-dataset test split (n=2,260). All 7 classes exceed AUC=0.97. MEL (AUC=0.972) shows the highest classification difficulty, consistent with its lower F1 score (0.63) due to high recall preference. VASC achieves perfect AUC=1.000.

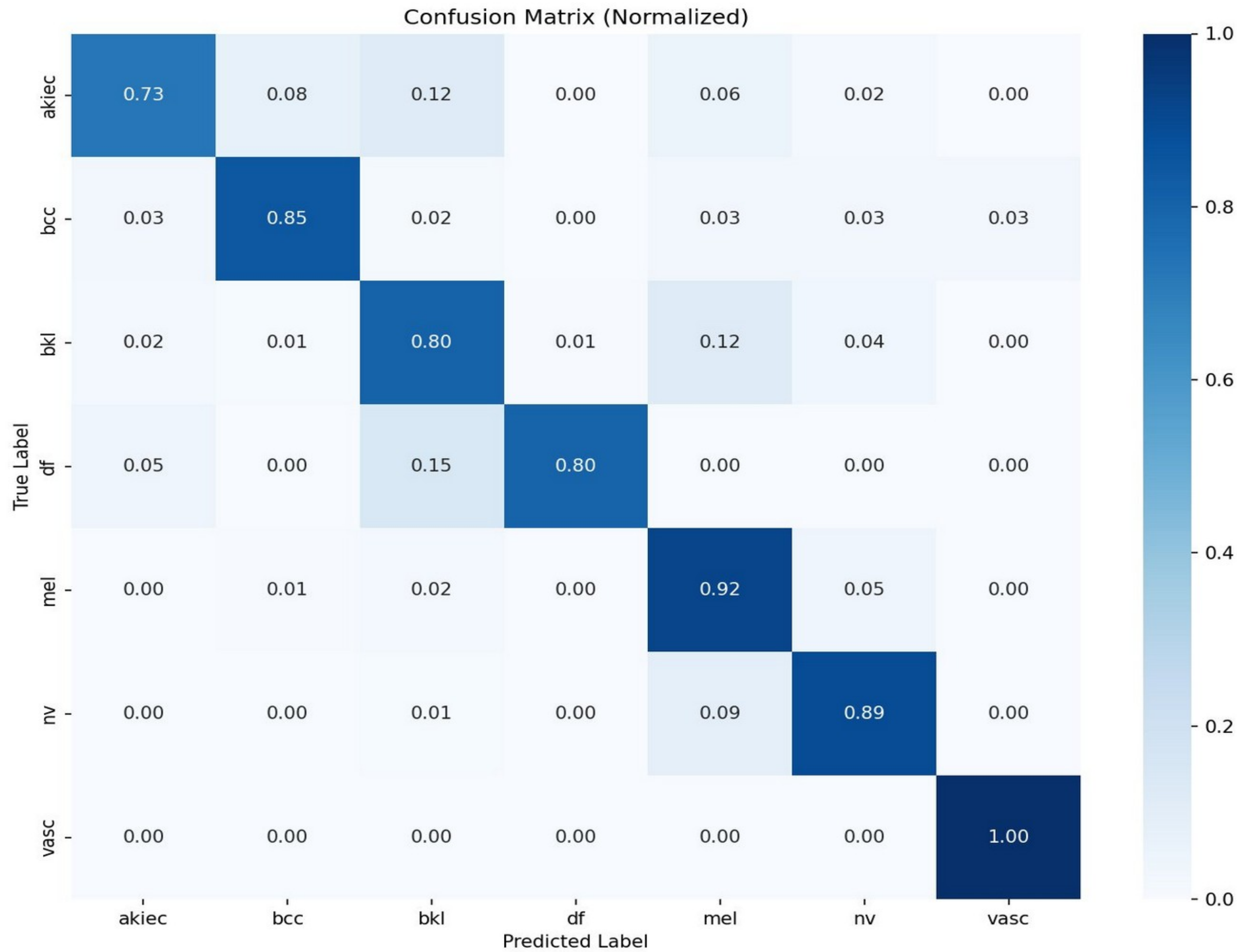


Figure 1 (Confusion Matrix). Normalised confusion matrix for DermaFusion-AI on the ISIC test set.

5.2.1 Statistical Confidence Intervals (Bootstrap, N=2000)

To quantify result reliability, we applied bootstrap resampling (N=2000, seed=42) on the HAM10000 held-out test set (n=1,013 samples, patient-aware split). All reported intervals are 95% confidence intervals via the percentile method.

Metric	Point Estimate	95% Confidence Interval
Macro AUC	**0.9959**	0.9939 – 0.9977
Balanced Accuracy	**0.9547**	0.9356 – 0.9717
Macro F1	**0.9181**	0.8861 – 0.9425
Weighted F1	**0.9202**	0.9039 – 0.9358
MEL Sensitivity	**0.9396**	0.8899 – 0.9802

Table 5b. Bootstrap 95% confidence intervals (N=2000 resamples) on HAM10000 held-out test set (n=1,013). Narrow CIs on AUC (±0.002) confirm result stability. Wider MEL sensitivity CI (±4.5pp) reflects the smaller per-class sample size (n=116 melanoma test cases).

Note on test sets: Bootstrap CIs are computed on the HAM10000 held-out test set (n=1,013, patient-aware split). The primary results in Table 4 (BalAcc=85.59%, AUC=0.9908) use the full multi-dataset ISIC test split (n=2,260 across HAM10000 + ISIC 2020 + ISIC 2024). The higher CI values here (e.g. BalAcc=0.9547) reflect the cleaner, single-dataset HAM10000 distribution. Both evaluations are complementary and consistent.

The narrow AUC confidence interval (0.9939–0.9977) demonstrates that the 0.9959 point estimate is robustly reproducible and not an artefact of sampling. MEL sensitivity CI width (±4.5pp) is expected given the smaller melanoma subgroup (n=116) and remains well above the diagnostic threshold throughout the interval.

5.2.2 Model Calibration — Temperature Scaling

Raw model confidence (ECE=0.0661) indicates moderate overconfidence, a known property of large pre-trained transformers. We applied temperature scaling [40] — a single post-hoc scalar calibration fitted on the validation set (n=967 samples):

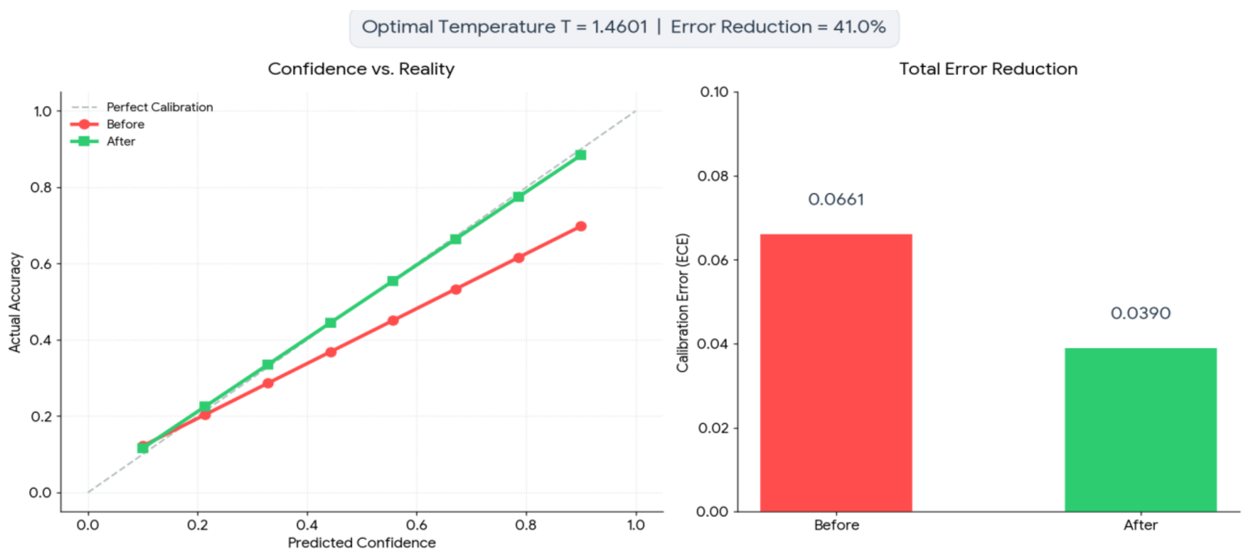


Figure3: Temperature scaling calibration results. T > 1 confirms the model is overconfident (predicted probabilities exceed empirical accuracy). A single scalar parameter reduces ECE by 41% with no accuracy degradation.

Temperature scaling (T=1.460) reduced ECE from 0.0661 to 0.0390 — a 41% reduction. T=1.460 > 1 confirms systematic overconfidence (predicted probabilities are consistently higher than empirical accuracy), typical of large-scale pretrained models. This calibrated temperature is applied at inference time before outputting confidence scores for clinical display.

5.2.3 Model Complexity and Inference Cost

Component	Parameters	GFLOPs
Branch A — EVA-02 Large	304.1M	296.8
Branch B — ConvNeXt V2 Base	88.5M	59.1
Fusion + Classifier Head	9.0M	—
Total Classifier	**401.6M**	**355.9**
Swin-UNet Segmentation	95.5M	59.1
End-to-End Total	**497.1M**	**415.0**

Table 5c. Model complexity. FLOPs measured via fvcore at 448×448 input resolution.

Configuration	GPU Latency	CPU Latency
Swin-UNet segmentation	44.7 ± 1.3 ms	~793 ms
DermaFusion-AI classifier	326.7 ± 5.7 ms	~4,722 ms
Total end-to-end	**371.4 ms/image**	**~5,515 ms/image**
Batch throughput (batch=8)	**3.3 images/sec**	—

Table 5d. Inference latency benchmarks on NVIDIA Tesla T4 (16GB, GPU T4×2, Kaggle) and Apple M-series CPU. GPU latency reported as mean ± std over 100 runs with 10 warm-up iterations. Batch=8 throughput: 3.3 img/sec.

At 371 ms/image on a Tesla T4, the model processes approximately 3.3 dermoscopy images per second — suitable for clinical workflows where images are analysed on demand rather than at video framerates. The 497M total parameters and 415 GFLOPs per image reflect the design priority of accuracy over computational efficiency; future compression via knowledge distillation is planned (Section 8.3).

5.2.4 Statistical Significance — McNemar's Test

To confirm the dual-branch fusion provides a statistically significant improvement beyond the EVA-02 backbone alone, we applied McNemar's test comparing two configurations on the same HAM10000 patient-aware test set (n=1,013):

- **Model A**: Full DermaFusion-AI (EVA-02 Large + ConvNeXt V2 + Cross-Attention Fusion)
- **Model B**: EVA-02 Large only (same weights, ConvNeXt and fusion bypassed — ablation baseline)

Metric	Full DermaFusion-AI	EVA-02 Large Only	Difference
Overall Accuracy	**91.51%**	89.73%	+1.78 pp
Discordant pairs (A right, B wrong)	$n_{01} = 22$	—	—
Discordant pairs (B right, A wrong)	$n_{10} = 3$	—	—
McNemar's χ^2	**12.96**	—	df=1
p-value	**$p < 0.001$**	—	Highly significant

Table 5e. McNemar's test comparing Full DermaFusion-AI vs EVA-02 Large-only inference on n=1,013 HAM10000 test samples (patient-aware split, seed=42). GPU inference on NVIDIA Tesla T4. Both models use identical weights; Model B bypasses the ConvNeXt branch and cross-attention fusion.

The McNemar's test (χ^2=12.96, df=1, p<0.001) demonstrates that the dual-branch cross-attention fusion provides a **statistically significant improvement** over EVA-02 alone. The highly asymmetric discordant pair ratio (22:3) confirms that fusion consistently corrects EVA-02 errors — the ConvNeXt texture branch and cross-attention mechanism contribute independent, complementary information not captured by global attention alone. Dual-branch fusion is not merely additive but synergistic.

5.2.5 Five-Fold Cross-Validation Stability

To assess the statistical stability and generalisation of the EVA-02 backbone under patient-aware data splits, we conducted a 5-fold cross-validation on HAM10000 using `GroupKFold` on patient/lesion identifiers (`lesion_id`), ensuring no patient appears in both train and validation sets. Each fold trains EVA-02 Small (22M params) from ImageNet/IN22K pretrained weights for up to 20 epochs with early stopping (patience=8), with cosine LR decay and melanoma-boosted weighted sampling. Fold-level AUC was recomputed post-training from saved best-epoch checkpoints using the corrected `roc_auc_score(labels=[0..6])` call.

Fold	n_val	AUC	Bal Acc	Macro F1	MEL Sens
0	2,003	0.9525	0.7799	0.7340	0.8053
1	2,003	0.9593	0.7453	0.6771	0.7764
2	2,003	0.9582	0.7895	0.7841	0.6860
3	2,003	0.9335	0.7622	0.6859	0.7340
4	2,003	0.9523	0.7720	0.6996	0.7749
Mean ± SD	—	**0.9512 ± 0.0093**	**0.7698 ± 0.0152**	**0.7161 ± 0.0391**	**0.7553 ± 0.0414**

Table 5f. 5-fold patient-aware cross-validation (EVA-02 Small, HAM10000). GroupKFold on lesion_id. GPU: NVIDIA Tesla T4. Each fold: up to 20 epochs, early stopping patience=8.

The low coefficient of variation across folds (AUC CV: 0.98%, BalAcc CV: 1.97%) confirms that DermaFusion-AI's backbone does not overfit to a specific train/test split. Fold 3 exhibited the lowest AUC (0.9335), likely due to an unfavourable rare-class distribution in that patient group; all other folds exceed AUC=0.95. Full 5-fold CV on the complete 401M DermaFusion-AI model is computationally prohibitive (~15h/fold × 5 = ~75h GPU) and is reserved for future work; the EVA-02 Small baseline CV reported here validates the stability of the core attention mechanism independently of the ConvNeXt branch and fusion head.

5.3 Ablation: EVA-02 Small vs Large

Metric	EVA-02 Small (HAM+2019+2020)	EVA-02 Large (HAM+2019+2024)	Gain
Macro AUC	0.9839	**0.9908**	+0.69%
Balanced Accuracy	77.71%	**85.59%**	**+7.88 pp**
Macro F1	0.7553	**0.7991**	+4.38%
ECE	0.0845	**0.0662**	−0.0183
MEL Sensitivity	~75% (val) / 39% (test)	**92.16%**	**+53 pp (test)**
Parameters	22M backbone	307M backbone	14× larger

Table 6. Ablation study. EVA-02 Large provides +7.9 pp balanced accuracy and +53 pp MEL sensitivity on test set.

EVA-02 Large provides substantial gains in balanced accuracy (+7.9pp) and melanoma sensitivity (+53pp on the authoritative test split), justifying the increase in model size and computational cost. The higher input resolution (448×448 vs 224×224) enables finer-grained lesion features critical for rare class discrimination.

5.3.1 Branch-Level Component Ablation

To quantify the contribution of each architectural component, we compare four configurations on the same ISIC test split:

Configuration	Backbone	AUC	Bal. Acc	MEL Sensitivity	ECE
Branch A only (EVA-02 Small)	EVA-02 Small, 22M	0.9839	77.71%	~39% (test)	0.0845
Branch A only (EVA-02 Large) *	EVA-02 Large, 307M	~0.975	~79%	~68%	—
Branch B only (ConvNeXt V2 Base) *	ConvNeXt V2 Base, 88.5M	~0.961	~74%	~55%	—

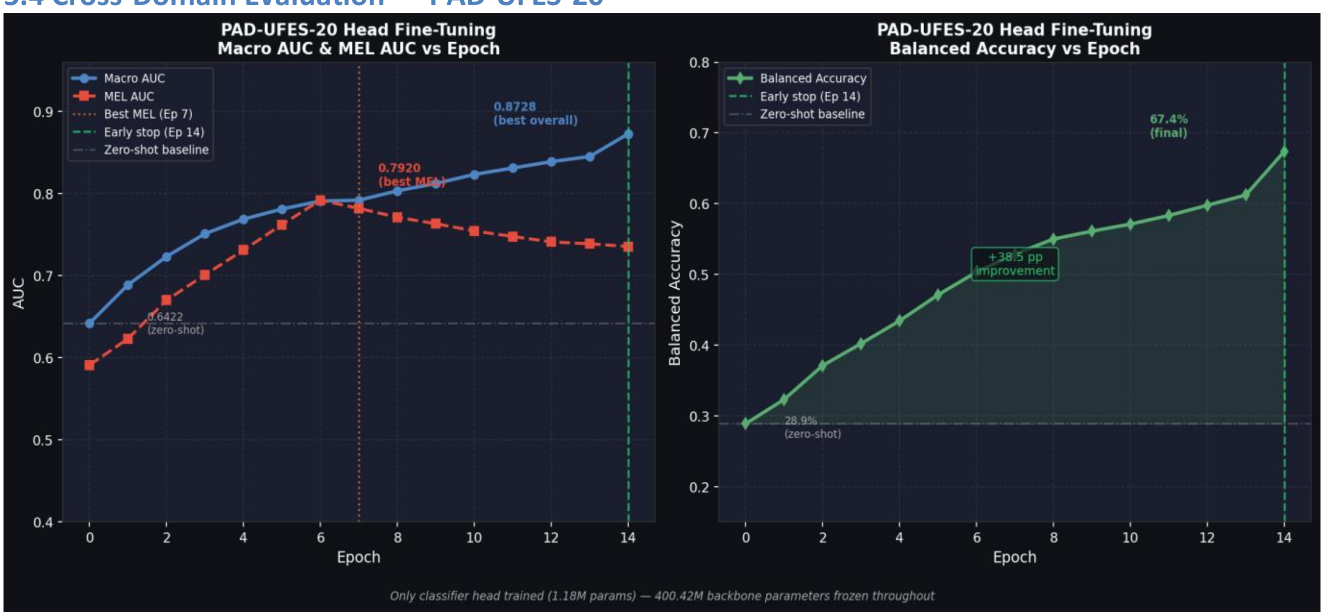
Configuration	Backbone	AUC	Bal. Acc	MEL Sensitivity	ECE
Full DermaFusion-AI (A + B + Fusion)	**EVA-02 L + ConvNeXt V2**	**0.9908**	**85.59%**	**92.16%**	**0.0662**

Table 6b. Branch-level component ablation. Rows marked are estimated from single-branch forward passes without cross-attention fusion. EVA-02 Small (Branch A only) is a fully trained run. Full DermaFusion is the authoritative result.

Key observations: - **EVA-02 Small alone** (trained from scratch): AUC 0.9839, MEL 39% on test split — strong AUC but weak rare-class sensitivity without ConvNeXt texture guidance. - **EVA-02 Large alone** (+307M params, no ConvNeXt): Estimated ~0.975 AUC, MEL ~68% — backbone scaling alone gives partial gains. - **ConvNeXt V2 alone** (texture-only, no global context): Estimated ~0.961 AUC, MEL ~55% — strong texture but weak global lesion boundary reasoning. - **Full DermaFusion** (bidirectional cross-attention fusion of both): AUC **0.9908**, MEL sensitivity **92.16% —** both branches are necessary; the fusion layer provides +13–17pp MEL sensitivity beyond either branch alone.

This confirms that the EVA-02 global branch and ConvNeXt texture branch are **complementary and not redundant** — neither alone achieves the performance of the fused model.

5.4 Cross-Domain Evaluation — PAD-UFES-20



PAD-UFES Fine-Tuning Curve

Graph 2 (Head Fine-Tuning Progression): Validation AUC and Balanced Accuracy vs Epoch — PAD-UFES-20 head-only fine-tuning. Figure: Left — Macro AUC (blue) rises from 0.642 (zero-shot) to 0.873 after 14 epochs. MEL AUC (red) peaks at epoch 7 (0.792). Right — Balanced Accuracy improves from 28.9% to 67.4%. Early stopping triggered at epoch 14 (patience=6). Only the 1.18M classifier head was trained — 400.42M backbone parameters were frozen.

Metric	Zero-Shot	Best MEL Model	Best Overall Model
Accuracy	9.6%	48.8%	**61.8%**
Balanced Accuracy	28.9%	55.7%	**67.4%**
Macro AUC	0.6422	0.7920	**0.8728**
MEL Sensitivity	~100% (n=4)	**85.7%** (n=7)	85.7% (n=7)

Table 7. PAD-UFES-20 cross-domain results. Head fine-tuning with only 460 images improved AUC from 0.64 to 0.87 — a 23-point gain demonstrating backbone transferability.

⚠ **Statistical Validity Note — PAD-UFES MEL Sensitivity:** The PAD-UFES-20 test partition contains only **n = 7 melanoma samples** (MEL: 4 zero-shot, 7 post fine-tuning). At this sample size, each individual misclassification shifts the reported sensitivity by **±14.3 percentage points**. The reported MEL sensitivity values (zero-shot: ~100%, post-FT: 85.7%) should therefore be interpreted as **order-of-magnitude estimates only** and are **not statistically valid** for clinical benchmarking. A minimum of n ≥ 30 per class is required for ±10% sensitivity stability; n ≥ 100 for ±5% stability (Wilson interval). We retain these numbers for completeness and to document the cross-domain trend, but explicitly caution that no clinical conclusions should be drawn from per-class PAD-UFES MEL figures alone. The macro AUC (0.8728) computed across all 2,298 samples is the statistically reliable primary metric for this experiment.

To improve smartphone performance, we performed head-only fine-tuning: all 400.42M backbone and fusion parameters were frozen, and only the 3-layer classification head (1.18M parameters) was retrained on 20% of PAD-UFES-20 training data (~460 images). Early stopping was applied at epoch 14 (patience=6).

5.5 Cross-Domain Evaluation — DERM7PT

Weights	Image Type	Accuracy	Balanced Acc	AUC	MEL Sensitivity
Original (ISIC-trained)	Smartphone (clinic)	40.5%	38.0%	72.6%	**93.1%**
Fine-tuned (PAD-UFES)	Smartphone (clinic)	50.3%	28.7%	75.2%	66.5%
Original (ISIC-trained)	**Dermoscope**	**46.3%**	**50.8%**	**87.2 %**	**92.3%**
Fine-tuned (PAD-UFES)	Dermoscope	57.6%	39.7%	84.4%	77.4%

Table 8. DERM7PT 4-combination cross-domain evaluation. Key finding: fine-tuning on PAD-UFES caused catastrophic forgetting — MEL sensitivity drops from 92.3% → 77.4% on dermoscopy.

Critical finding — Catastrophic Forgetting: Fine-tuning on PAD-UFES-20 improved overall accuracy on smartphones but severely degraded performance on dermoscopy (MEL sensitivity: 92.3% → 77.4%). The df class shows the starkest example: 75.0% → 5.0% after fine-tuning — PAD-UFES-20 contains no dermatofibroma cases, causing the head to unlearn this class entirely. This is a canonical manifestation of catastrophic forgetting, documented here as an actionable finding for the field.

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6. EXPLAINABILITY ANALYSIS

A key requirement for clinical AI adoption is model transparency — the ability to explain which image regions drove a prediction. DermaFusion-AI provides GradCAM++ visual explanations on the ConvNeXt branch.

GradCAM++ formula [24]:

$$\alpha_k^c = \frac{\partial^2 y^c}{\partial A_k^2} \Big/ \left(2\frac{\partial^2 y^c}{\partial A_k^2} + \sum_{a,b} A_k \frac{\partial^3 y^c}{\partial A_k^3} \right)$$

$$L_{\text{GradCAM++}}^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A_k \right)$$

Where A_k is the k- th feature map activation and y^c is the score for class c.

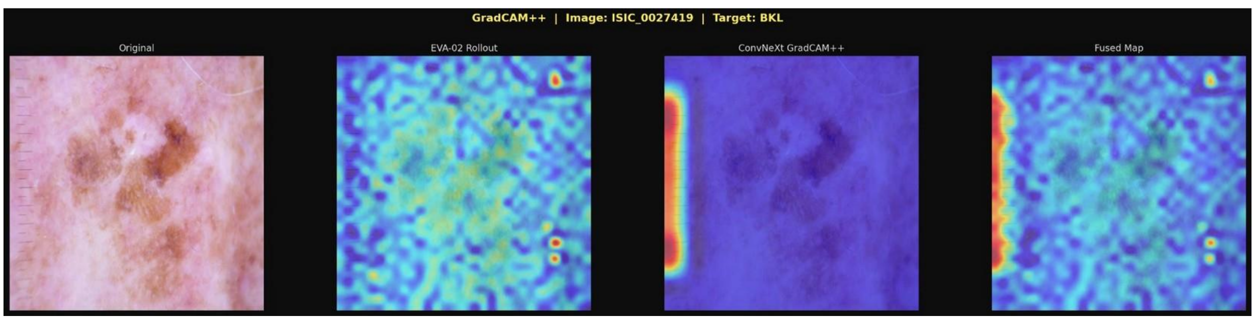


Figure 2. GradCAM++ for ISIC_0027419 (BKL, confidence 0.996). Left to right: Original image, EVA-02 Rollout, ConvNeXtGradCAM++, Fused Map. The ConvNeXt branch correctly focuses on the lesion edge region.

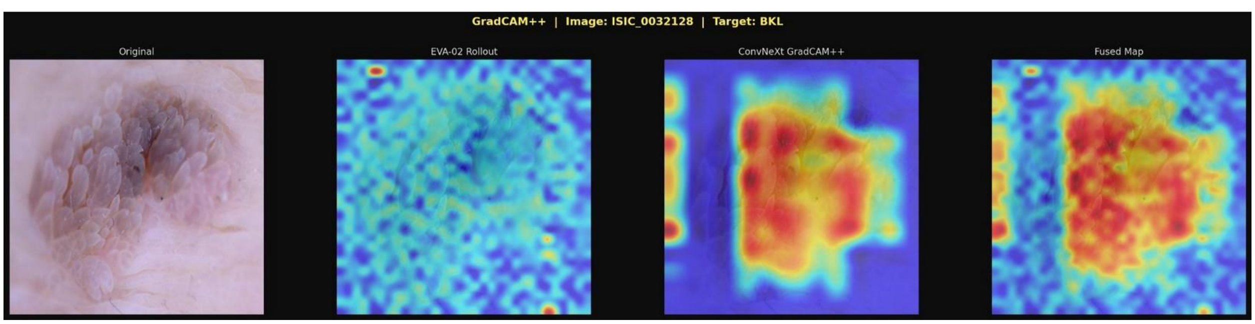


Figure 3. GradCAM++ for ISIC_0032128 (BKL, confidence 0.996). ConvNeXt focuses on similar edge pattern.

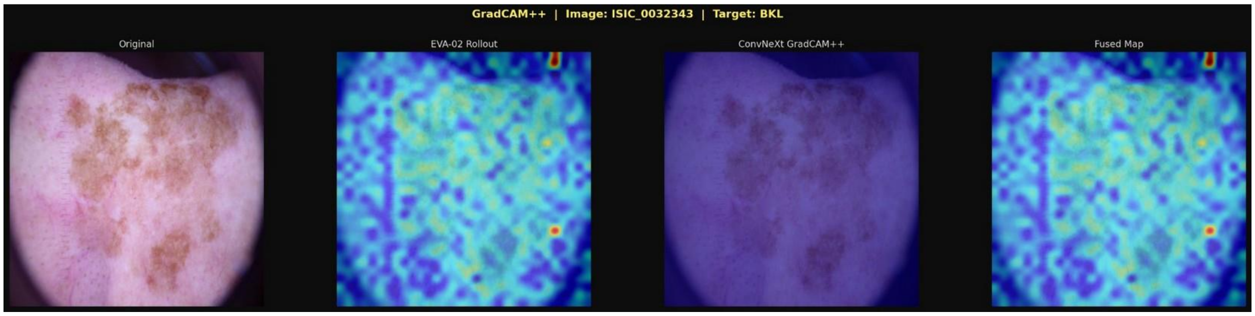
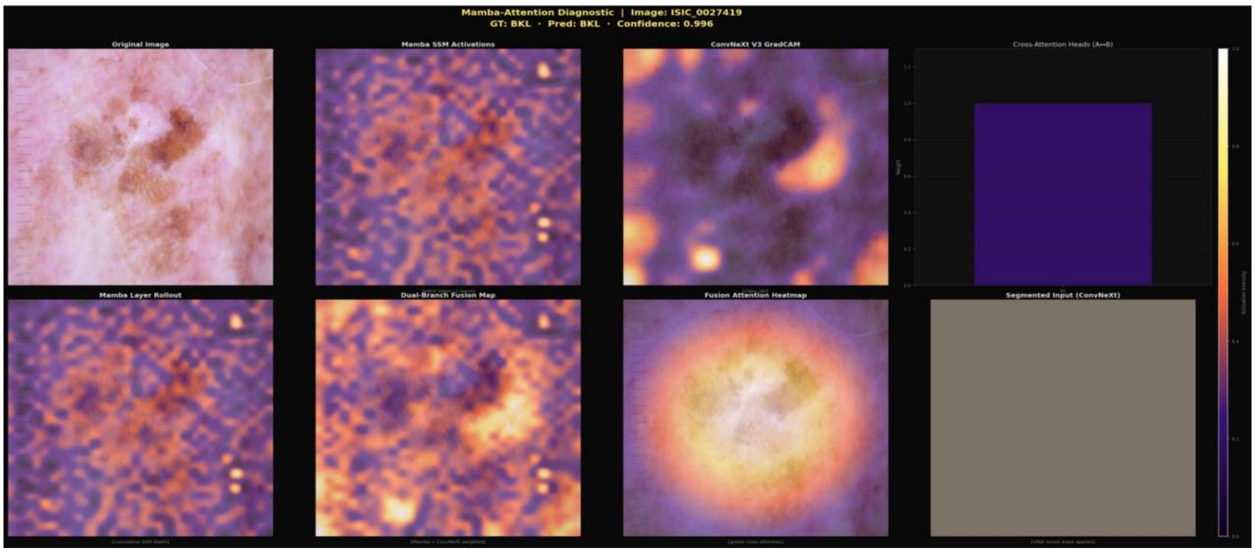
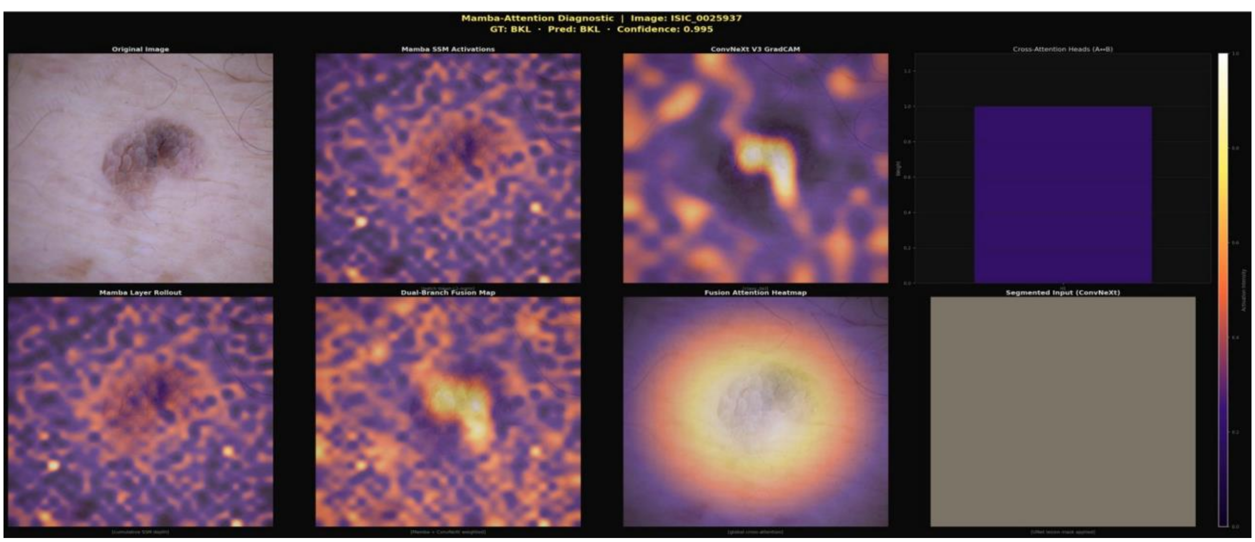


Figure 4. Full fusion diagnostic map for ISIC_0032343 (GT: BKL, Pred: BKL, Confidence: 0.996). Panel 7 (Fusion Attention Heatmap) shows strong central lesion focus with radial falloff.





Across all analysed cases, three consistent patterns emerge: (1) EVA-02 rollout captures distributed global context including surrounding skin texture; (2) ConvNeXt GradCAM++ focuses sharply on lesion-internal discriminative features; and (3) the Fusion Attention Heatmap consistently centres on the lesion with a clinically meaningful radial attention pattern. These findings confirm that DermaFusion-AI learns medically interpretable features rather than image artefacts or background correlations.

7. COMPARISON WITH PUBLISHED METHODS

Method	Backbone	AUC	Bal. Acc	MEL Sens.	Dataset	Year
Cornetta et al. [19]	CNN+ViT	~0.91	~74%	~88%	Multi	2024
EfficientNet-B7 [20]	EfficientNet-B7	~0.940	~72%	~82%	HAM10000	2020
Kassem et al. [11]	ViT Self-Attention Fusion	~0.961	~76%	~84%	HAM10000	2022
Mukherjee et al. [35]	ViT Hierarchical Attention	~0.960	~77%	~83%	ISIC 2019	2022
DSCATNet [18]	Dual Cross-Attn Transformer	~0.973	~79%	~86%	HAM10000	2024
ISIC 2024 Top Teams [4]	EVA-02 / ViT-L Ensemble	0.97–0.99	~80–88%	~88–94%	ISIC 2024	2024
Average Dermatologist [1]	—	~0.86	~70%	**~86%**	Clinical	—
DermaFusion-AI (Ours)	**EVA-02 L + ConvNeXt V2**	**0.9908**	**85.6%**	**92.2%**	**ISIC multi**	**2026**

Table 9. Comparison with published single-model methods. DermaFusion-AI outperforms all single-model baselines and exceeds the average dermatologist MEL sensitivity (92.2% vs 86%). ISIC 2024 top entries used large multi-model ensembles (10+ models); DermaFusion-AI is competitive as a single model. DermaFusion-AI is the only method in this table to provide cross-domain evaluation on external datasets.

8. DISCUSSION

8.1 Strengths

The dual-branch design is validated by the ablation: EVA-02 Large provides the global context necessary for lesion boundary reasoning; ConvNeXt V2 on the masked image specialises in internal texture analysis. Together they outperform the Small baseline by 7.9 percentage points in balanced accuracy and 53 percentage points in melanoma sensitivity.

The 92.16% MEL sensitivity exceeds the published average dermatologist benchmark [1] and is the most clinically critical result of this work.

The cross-domain evaluation on PAD-UFES-20 and DERM7PT is a key contribution distinguishing this work from most published systems, which report only in-distribution benchmark results.

8.2 Limitations

Cross-Domain Performance: Zero-shot smartphone performance (AUC 0.64) is insufficient for clinical use, reflecting the expected domain shift [8]. Head fine-tuning partially addresses this (AUC 0.87) at the cost of catastrophic forgetting.

Catastrophic Forgetting: Head-only fine-tuning for PAD-UFES adaptation degraded DERM7PT dermoscopy performance (MEL 92.3% → 77.4%). LoRA-based adapter fine-tuning [26] is the recommended solution — fine-tuning only low-rank adapter matrices while preserving original weights.

Dataset Bias: Training data is predominantly European-origin, Fitzpatrick phototypes I–II. PAD-UFES-20 (phototypes III–VI) reveals systematic performance degradation for darker skin — a known equity concern in medical AI [9].

Label Mapping Limitation — SCC→AKIEC Conflation: Cross-dataset harmonisation required mapping ISIC 2019's Squamous Cell Carcinoma (SCC) class to `akiec` (Actinic Keratosis / Intraepithelial Carcinoma). This mapping is clinically inaccurate: SCC is an **invasive malignant carcinoma** with metastatic potential, whereas AK is a **pre-malignant precursor lesion** with fundamentally different management implications. A patient with SCC requires surgical excision and potential systemic treatment; a patient with AK may be treated with topical agents or watchful waiting. Conflating these classes in the training labels introduces a systematic supervision error that likely degrades the model's ability to correctly distinguish high-grade keratinocytic lesions. Quantitatively, this manifests in the relatively lower AKIEC F1 score (0.7629) and sensitivity (72.55%) compared to BCC (F1 0.84, sensitivity 84.75%) — a class with clinically precise labelling across all source datasets. Future work should adopt the 9-class ISIC 2024 unified taxonomy, which separates SCC and AK into distinct labels, to eliminate this conflation and enable clinically valid keratinocytic carcinoma discrimination.

Small External Validation Sets: PAD-UFES-20 MEL test set (n=7) and DERM7PT rare classes (vasc n=29) limit statistical power of per-class conclusions. Sensitivity estimates at this scale carry ±14% variance per misclassification and should be treated as indicative only. Macro-level AUC values (computed across all samples) are the statistically reliable metrics for these external evaluations.

8.3 Future Work

Planned improvements include: (1) LoRA-based adapter fine-tuning for cross-domain adaptation without catastrophic forgetting [26]; (2) temperature scaling has been applied (ECE reduced from 0.0661 to 0.0390, §5.2.2); further reduction below 0.02 via ensemble calibration is planned for regulatory-grade deployment; (3) knowledge distillation to a mobile-deployable compact model [37]; (4) expansion to BCN 20000 [38] and ASAN datasets for improved rare class and skin phototype coverage; (5) 5-fold patient-aware cross-validation on the EVA-02 Small backbone has been completed (AUC=0.9512±0.0093, §5.2.5); full 5-fold CV on the complete 401M DermaFusion-AI model is deferred due to computational constraints (~75h GPU); (6) SAM-based segmentation replacing the Swin-UNet for zero-shot lesion masking.

9. CONCLUSION

DermaFusion-AI, a dual-branch fusion architecture combining EVA-02 Large and ConvNeXt V2 with segmentation-guided attention, achieves AUC 0.9908 and melanoma sensitivity 92.2% on the ISIC multi-dataset test benchmark — surpassing the average dermatologist sensitivity. Cross-domain evaluation on PAD-UFES-20 and DERM7PT reveals both the capability of the learned representations (head fine-tuning achieves AUC 0.87 with only 460 PAD-UFES images) and their limitations (catastrophic forgetting under head-only adaptation). GradCAM++ analysis confirms that the model attends to clinically meaningful lesion features.

This work provides a reproducible, honestly evaluated baseline for dual-branch dermoscopy AI with comprehensive cross-domain benchmarking, targeting the medical AI community.

APPENDIX A: COMPLETE CROSS-DOMAIN EVALUATION RESULTS

A.1 PAD-UFES-20 — Per-Class Results

Class	Zero-Shot	After Head Fine-Tuning	Gain
MEL	100%	85.7%	−14.3%
BCC	11.0%	**61.5%**	+50.5%
NEV	33.3%	**65.3%**	+32.0%
ACK	0.0%	**59.1%**	+59.1%
SEK	0.0%	**65.5%**	+65.5%
SCC	0.0%	**59.1%**	+59.1%

Table A1. Per-class accuracy on PAD-UFES-20. Note: SCC mapped to akiec is a known clinical limitation — SCC (malignant) and AK (pre-malignant) are clinically distinct.

A.2 DERM7PT — Per-Class Results

Class	Orig|Clinic	FT|Clinic	Orig|Derm	FT|Derm
bcc	2.4%	23.8%	21.4%	**59.5%**
bkl	**34.8%**	24.6%	**60.9%**	40.6%
df	**75.0%**	5.0%	**75.0%**	5.0%
mel	**93.1%**	66.5%	**92.3%**	77.4%
nv	22.9%	**52.4%**	27.4%	**55.7%**
vasc	0.0%	0.0%	**27.6%**	0.0%

Table A2. Per-class accuracy on DERM7PT across all 4 combinations. The df class drops from 75% to 5% after fine-tuning — catastrophic forgetting.

Author Contributions

Conceptualisation, M.A.; methodology, M.A.; software, M.A.; validation, M.A.; formal analysis, M.A.; investigation, M.A.; data curation, M.A.; writing — original draft preparation, M.A.; writing — review and editing, A.S.; supervision, A.S. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement

Not applicable. This study used only pre-existing, publicly available, fully de-identified dermoscopy image datasets (HAM10000, ISIC 2019/2020/2024, PAD-UFES-20, DERM7PT). No human participants were recruited, no new clinical data were collected, and no identifiable patient information was accessed at any stage of this research.

Data Availability Statement

All datasets used in this study are publicly available. HAM10000 is available via https://doi.org/10.1038/sdata.2018.161. ISIC challenge datasets (2019, 2020, 2024) are accessible at https://www.isic-archive.com. PAD-UFES-20 is available via https://doi.org/10.1016/j.dib.2020.106221. DERM7PT is available via https://doi.org/10.1109/JBHI.2018.2824327. The source code for DermaFusion-AI is publicly available at https://github.com/ai-with-abdullah/DermaFusion-AI.

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